ScanCrypt

Post-Quantum Cryptographic Security Assessment

Project Name	m4fstack
Analysis Date	2026-07-09 20:45:46
ScanCrypt Version	1.0
Algorithm	ML-KEM / Kyber
Security Level	Unknown ML-KEM Variant
Implementation Type	Masked Implementation

High

Confidentiality Notice

This document contains confidential assessment details based on ScanCrypt analysis. Unauthorized disclosure is prohibited.

Report generated automatically by ScanCrypt.

Executive Summary

Overall Risk	High	**Algorithm**	ML-KEM / Kyber
Security Level	Unknown ML-KEM Variant	**Implementation Type**	Masked Implementation
Programming Language	C/C++	**Build System**	Not detected
Files	3	**Source Files**	2
Header Files	1	**Lines of Code**	906
Rules Executed	15	**Rules Passed**	9
Rules Failed	6	**Analysis Confidence**	99
Detection Confidence	96%	**Analysis Duration**	00:01:25

Critical: 0

High: 1

Medium: 0

Low: 0

Information: 0

Project Inspection

Algorithm	ML-KEM / Kyber
Security Level	Unknown ML-KEM Variant
Implementation Type	Masked Implementation
Programming Language	C/C++
Build System	Not detected
Files	3
Source Files	2
Header Files	1
Lines of Code	906
Detection Confidence	96%

Detection Evidence

- KYBER_N detected in poly.h
- KYBER_Q detected in poly.c
- KYBER_K detected in poly.c
- poly_compress() detected in poly.h
- poly_decompress() detected in poly.h
- ntt() detected in poly.h
- invntt() detected in poly.h

Findings Summary

Rule ID	Severity	Category	Status	File	Line
IMP-001	High	Configuration	Failed	poly.c	42

Detailed Findings

IMP-001 — High

KYBER_K parameter disclosure

Rule ID	IMP-001
Severity	High
Category	Configuration
Affected Algorithm	ML-KEM / Kyber
Affected File	poly.c
Affected Line	42

Description
KYBER_K parameter was detected but not validated against expected values.

Potential Security Impact
Incorrect parameter values can reduce algorithmic security.

Technical Explanation
Kyber security level depends directly on KYBER_K. The extracted KYBER_K value was missing or malformed.

Research-based Mitigation Recommendation
Ensure KYBER_K is defined with a supported integer value of 2, 3, or 4.

References
NIST PQC Kyber specification

RFC 9420

Rule Execution Summary

✗ IMP-001 — Failed

Statistics

Total rules executed
15

Passed rules
9

Failed rules
6

Analysis duration
00:01:25

Files analysed
3

Source lines analysed
906

Execution Progress

Passed: 9 / 15

Failed: 6 / 15

Appendix

Analysed Files

poly.c

poly.h

main.c

Tool version
1.0

Report generated
2026-07-09 20:45:46

Scan configuration
Default ScanCrypt profile

Supported algorithms
ML-DSA (Dilithium), ML-KEM (Kyber)

Supported implementation types
Portable C / Reference, AVX2 Optimized, ARM Cortex-M / Embedded, Masked Implementation